Penna
A Tool for Collecting **Qualitative Textual Data**

Penna is a web application for collecting self-administered writings for research purposes. It was built at Finnish Social Science Data Archive (FSD) and has been in production since 2017. FSD offers Penna as a service for researchers to collect metadata and writings easily. FSD gets to archive the collected data.

FSD and the researcher construct together a questionnaire which is a web form to be shared with possible respondents. The respondents fill and submit the form. After the collection period is over FSD can export the questionnaire and writings as packages that can be delivered to themselves and the researcher.

Features

DDI export
- The DDI file consists of questions/variables of the questionnaire and metadata about the collection
- The file is compatible with DDI-C versions 2.1 and 2.5
- It can be used as a basis for documenting the data

Respondents must state if they agree to let FSD archive their responses.

Data export
- Two types of data exports:
  - One that contains all responses is for the researcher
  - Another that contains only the responses that will be archived at FSD
- A DDI Codebook of the questionnaire is included in the FSD export package

Questionnaire
- Customizable
- Open for a predetermined duration
- Can be public or private
  - Public questionnaires are listed on the front page
  - A private questionnaire can be accessed only if the respondent knows the URL of the questionnaire

The respondent can continue filling the questionnaire after first submission.

Questionnaire export and import
- The questionnaire can be exported as json
- The exported questionnaires can be later imported to create new questionnaires

Technical details

- Built with Python 3 using Pyramid web framework
- Jinja2 as templating language
- Responsive front-end using Foundation
- PostgreSQL used for storage
- SQLAlchemy for ORM
- Shibboleth authentication for admin functionalities
  - Public site can be accessed by everyone
  - Only authenticated users can access admin site
- AMQP messages from life-cycle events for integration to other systems
- The DDI file is generated using the lxml Python library

Benefits for different users

Researcher
- Gets to collect data for research
- Gets help with collecting data and archiving, free of charge
- Penna provides a simple path to archiving collected data

Respondent
- Gets to participate in data collection

FSD
- Gets the collected data in structured forms (including as a basis for documenting data)
- Penna adds marketing value

Check out the demo video on YouTube

https://youtu.be/c_8JiGdJFtY

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